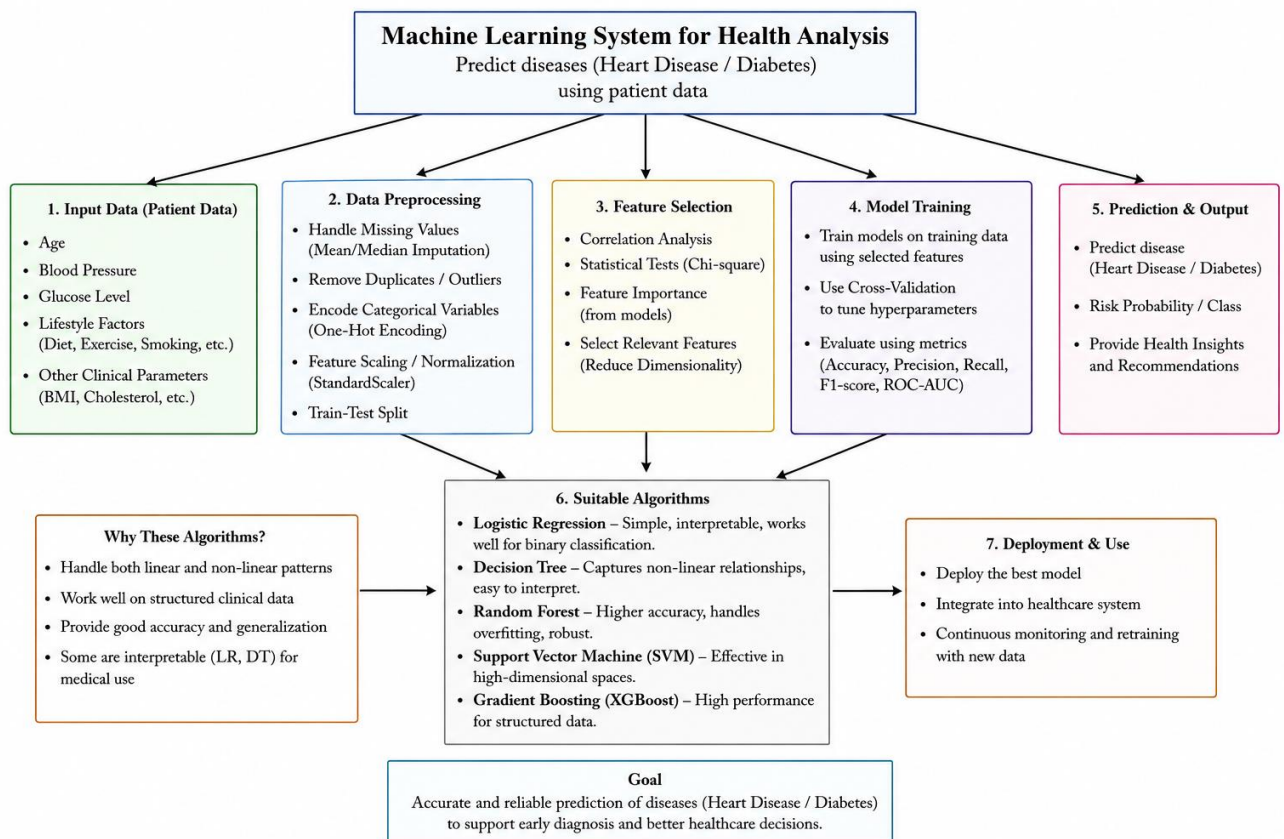


ITA0609 – Machine Learning

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Machine Learning System for Health Analysis



Find-S Algorithm

Find-S Algorithm - Loan Approval Dataset

Problem

Learn the most specific hypothesis that is consistent with all positive examples (Loan = Yes) from the dataset.

Dataset

ID	Employment Status	Salary Level	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clear	Yes
2	Self-Employed	Medium	Good	Clear	Yes
3	Employed	High	Excellent	Clear	Yes
4	Unemployed	Low	Poor	Default	No
5	Employed	Medium	Good	Clear	No
6	Self-Employed	High	Good	Default	No

Attributes

- Employment Status
{Employed, Self-Employed, Unemployed}
- Salary Level
{High, Medium, Low}
- Credit Score
{Excellent, Good, Poor}
- Loan History
{Clear, Default}

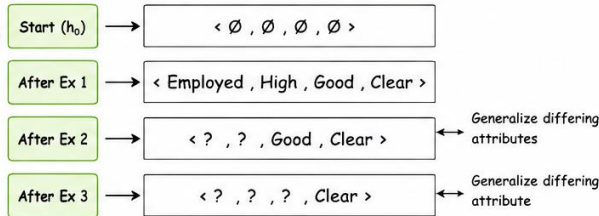
Find-S Algorithm

1. Initialize $h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$ (most specific)
2. For each positive example, generalize h to cover it.
3. Ignore negative examples.

Initialization

$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
(Most specific hypothesis)

Apply Find-S on Positive Examples



Final Hypothesis (Most Specific)

$h = \langle ?, ?, ?, \text{Clear} \rangle$

Means: Loan is approved if Loan History is Clear
(other attributes can be any value)

Key Points

- Find-S finds the most specific hypothesis.
- It only uses positive examples.
- It generalizes when a positive example is not covered by current hypothesis.
- ? denotes "any value".

Interpretation

The most specific hypothesis consistent with all positive examples is:

$h = \langle ?, ?, ?, \text{Clear} \rangle$

Only Loan History = Clear is required for loan approval in this dataset.

$\emptyset$ = most specific (no value)
? = any value